C. Anatomy of Calculated Metrics: Gene Architecture & Codon Physics

Biophysical basis of CDS GC%, GC3%, Whole-Gene GC%, Intron Compaction, and Direct Synonymous Shifts

1. Gene Architecture: Full Gene GC% vs. CDS GC% & Triplet Decomposition



Whole-Gene GC%: Exons + Introns + UTRs (Reflects regional chromatin & isochore baseline GC)

CDS GC%: Coding sequence only (Concatenated Exons 1, 2, 3; reflects amino acid & mutational bias)

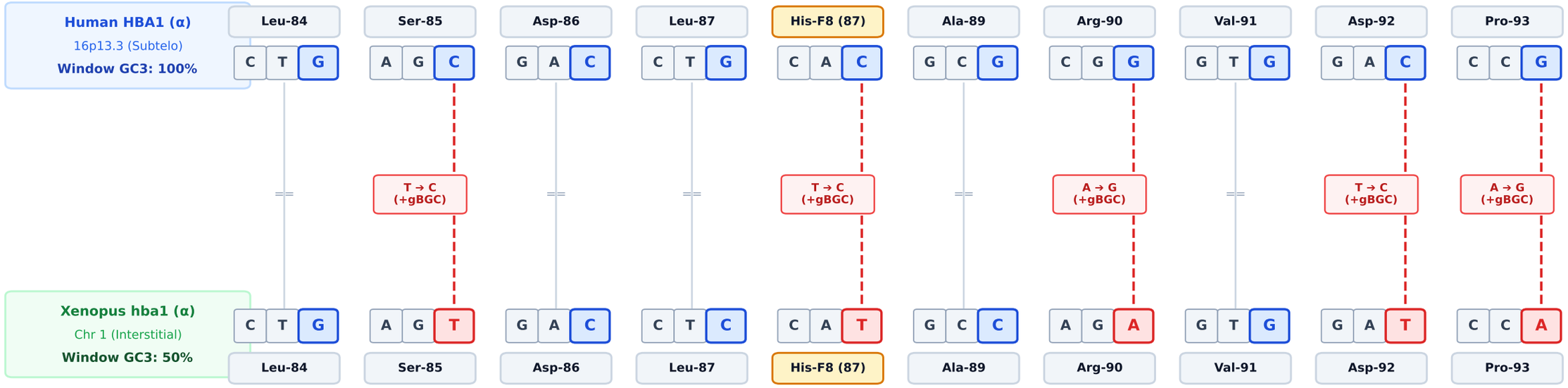
Codon Triplet Breakdown:

- GC1: Moderate purifying constraint (~50–65% GC)
- GC2: Locked in Purifying Corridor (45–55%)
- GC3: Synonymous site! Barometer of gBGC (35–98% GC)
- GC4d: 4-Fold degenerate (100% synonymous; pure gBGC)



2. Conserved Globin Heme Pocket: Proximal Histidine (His-F8) Motif & Synonymous GC3 Divergence

Comparison across 350+ Myr: 100% Identical Amino Acids, Yet 100% of Divergent 3rd Positions Shifted to G/C in Human!



3. Bernardi's Compaction Law & Telomeric Distance Driver

Bernardi's Compaction Law: High-GC isochores suppress intron expansion; low-GC accumulate massive introns.

Human HBA1 (α): 843 bp (Intron 2 = 140 bp) [Compact High-GC] vs. Human HBB (β): 1,608 bp (Intron 2 = 850 bp) [Expanded Low-GC]

Distance-to-Telomere (dtelo / Lchr) Drives gBGC:

- Meiotic crossovers are physically biased toward chromosome ends, accelerating GC-biased gene conversion (gBGC).
- **Subtelomeric loci (dtelo < 1 Mb) undergo runaway GC3 elevation; internal/centromeric loci remain buffered in AT-rich baselines.**

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Relative Telomeric Distance: dtelo_rel = min(Pos, Chr_Len - Pos) / Chr_Len    (0.000 Subtelomeric →  0.500 Metacentric)
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